Vertical Integration with Partial Ownership: Evidence from a Differentiated Products Market

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Abstract

This paper studies the competitive effects of vertical integration between upstream producers and downstream distributors in a differentiated-products market. While vertical integration can generate efficiency gains through improved coordination, it may also create incentives to disadvantage rival products distributed through shared downstream firms. Using a difference-in-differences design, I document that integration is associated with lower prices for products owned by integrating firms and higher prices for competing third-party products. I then estimate a structural demand model to recover brand-level substitution patterns, showing that consumers primarily substitute within formulation categories and across package formats, with relatively low cross-brand elasticities. Building on these estimates, I develop a vertical supply model that incorporates partial ownership and downstream pricing incentives. Counterfactual simulations indicate that foreclosure incentives play a quantitatively important role and can outweigh efficiency gains in shaping welfare outcomes. Overall, the results highlight the importance of accounting for partial ownership and downstream incentives when evaluating the welfare consequences of vertical integration.

Data Source and Disclaimer:

Researcher(s)' own analyses calculated (or derived) based in part on data from Nielsen Consumer LLC and marketing databases provided through the NielsenIQ Datasets at the Kilts Center for Marketing Data Center at The University of Chicago Booth School of Business.

The conclusions drawn from the NielsenIQ data are those of the researcher(s) and do not reflect the views of NielsenIQ. NielsenIQ is not responsible for, had no role in, and was not involved in analyzing and preparing the results reported herein.

To comply with data-use agreements, the analysis intentionally aggregates brands, obscures ownership structures, and suppresses specific institutional details. While informed readers may recognize similarities with known industries, the mapping between model entities and real-world firms is not one-to-one.

1 Introduction

Vertical integration has long occupied a central place in industrial organization and antitrust analysis. Theoretical work highlights two opposing forces associated with vertical integration. On one hand, integration between upstream and downstream firms can eliminate double marginalization, potentially lowering prices and increasing total surplus (Spengler, 1950). On the other hand, integration may generate foreclosure incentives, whereby integrated firms disadvantage rival upstream or downstream competitors through pricing or strategic conduct, potentially reducing competition and harming welfare (Asker and Bar-Isaac, 2014). The net effect of vertical integration therefore depends critically on industry structure, ownership arrangements, and the nature of competition.

This paper empirically studies the effects of vertical integration in a differentiated consumer goods industry characterized by a multi-tier supply chain. The industry features upstream producers that supply key inputs to downstream firms responsible for production and distribution. Historically, upstream producers have relied on partially owned or independently operated downstream partners, creating complex ownership and contracting relationships. During the study period, two major upstream firms completed vertical integration transactions with their principal downstream partners, while a third upstream firm continued to rely on shared downstream production arrangements. These events provide a natural setting to evaluate the pricing and welfare consequences of vertical integration under partial ownership and shared distribution.

To preserve confidentiality requirements imposed by data providers, firm and brand identities are anonymized throughout the paper. I refer to upstream producers as UP1, UP2, and UP3; downstream firms as B1, B2, and so forth; and products as X, Y, and Z. The analysis focuses on two vertical integration events in which UP1 and UP2 integrated with major downstream partners. These events occurred in different subperiods of the sample and affected a substantial share of downstream production capacity. Prior to integration, both UP1 and UP2 held significant minority ownership stakes in their downstream partners, while UP3 did not vertically integrate and instead relied on contractual downstream arrangements.

The empirical analysis begins with a difference-in-differences event study that examines changes in prices and market shares following integration. The results indicate that vertical integration is associated with modest and statistically insignificant price decreases for integrated products, with estimated reductions of 1.3 percent following the UP1 event and 0.69 percent following the UP2 event. In contrast, prices of rival products produced under shared downstream arrangements increase significantly, rising by 6.64 percent and 0.8 percent, respectively. These findings are consistent with the presence of both efficiency effects and foreclosure incentives.

To interpret these reduced-form results, I estimate a structural demand model to recover consumer substitution patterns and brand-level price elasticities. The estimated elasticities suggest that consumers primarily substitute within broad product categories, with median brand-level own-price elasticity around -4. Cross-brand substitution is present but more limited. Building on these demand estimates, I develop a vertical supply-side model that incorporates partial ownership, exclusive production territories, and shared downstream arrangements. This framework allows upstream firms to internalize downstream profits to varying degrees and captures the incentives created by partial integration.

The structural results indicate that pricing incentives generated by vertical integration may lead to soft foreclosure behavior, whereby integrated firms raise the prices of rival products distributed through shared downstream partners. Counterfactual simulations suggest that removing efficiency gains reduces total surplus by 0.687 percent, while removing foreclosure incentives increases total surplus by 1.2 percent. A full divestiture scenario leads to a substantial reduction in total surplus of 17.765 percent, driven largely by declines in consumer surplus. These results highlight the importance of accounting for partial ownership and foreclosure incentives when evaluating vertical mergers.

This paper contributes to several strands of the literature. First, it relates to empirical studies of vertical integration in differentiated product markets that combine demand estimation with structural supply-side modeling (Berto Villas-Boas, 2007; Crawford, Lee, Whinston and Yurukoglu, 2018; Yang, 2020). Second, it adds to the literature on consumer demand in this industry, which documents substitution patterns using retail scanner data (Dubé, 2005; Dhar, Chavas, Cotterill and Gould, 2005; Chan, 2006; Lopez, Liu and Zhu, 2015; Chen, Reinhardt and Syed Shah, 2022). Third, it complements recent reduced-form analyses of vertical integration that document price effects but abstract from partial ownership and downstream profit internalization (Luco and Marshall, 2020; Adachi, 2020).

Relative to this existing work, the key contribution of this paper is to explicitly model partial ownership and shared downstream production within a unified vertical framework. Unlike studies that treat vertical integration as a binary change in ownership, this paper allows for continuous ownership stakes and shows how partial alignment of incentives alters pricing behavior and welfare outcomes. By disentangling efficiency and foreclosure mechanisms, the analysis provides a more nuanced assessment of vertical integration and contributes to ongoing policy debates regarding the competitive effects of vertical mergers.

Finally, this paper relates to the theoretical literature on partial ownership in vertical relationships, which emphasizes how incomplete integration can alter firms' incentives and competitive outcomes (Levy, Spiegel and Gilo, 2018; Hunold, 2020). It also contributes to a growing empirical literature that allows for partial alignment of firms' objectives rather than assuming fully integrated or fully independent firms. For example, Miller and Weinberg (2017) study horizontal mergers in the beer industry by permitting non-binary ownership structures in the profit-maximization problem. In the context of vertical relationships, Crawford et al. (2018) and Cuesta, Noton and Vatter (2019) analyze integration in multichannel television and hospital–insurer markets, respectively, using Nash-in-Nash bargaining frameworks in which integrated firms maximize a weighted sum of their own and their partners' profits. While Crawford et al. (2018) model foreclosure incentives through an explicit foreclosure parameter, Cuesta et al. (2019) capture such incentives through profit differentials across contractual arrangements. Given the institutional environment studied in this paper, I abstract from bargaining and instead adopt a supply-side model in which firms internalize downstream profits according to observed ownership stakes, allowing partial integration to influence pricing incentives directly.

The remainder of the paper is organized as follows. Section 2 describes the market structure, Section 3 describes data, Section 4 describes some reduced-form analysis of the effects of vertical integration on product prices, Section 5 presents the demand-side model and elasticity estimates, Section 6 presents the supply-side model and results of producers marginal costs and markups, Section 7 presents the counterfactuals, and Section 8 concludes.

2 Background

2.1 Vertical Structure of the Industry

The supply chain studied in this paper features a two-tier vertical structure consisting of upstream producers and downstream distributors. Upstream firms supply branded intermediate inputs that are combined with local ingredients and packaging services by downstream distributors. Downstream distributors are responsible for final production activities—combining inputs with other ingredients—as well as packaging, transportation, and delivery to retail outlets. These downstream activities are economically significant, as they involve substantial logistical and merchandising costs.

A large share of products in this industry are delivered through decentralized distribution with in-store servicing rather than through centralized warehouses. Under this system, downstream distributors transport products directly to retail locations, stock shelves, and manage in-store displays. As a result, transportation and retail servicing costs are borne primarily by downstream firms rather than retailers, making downstream pricing incentives and cost structures central to competitive outcomes (Fry, Spector, Williamson and Mujeeb, 2011).

Two major upstream producers, denoted UP1 and UP2, each operate franchised downstream systems to facilitate production and distribution. These systems, labeled Group 1 and Group 2, consist of downstream firms that hold exclusive rights to distribute the upstream producer's products within predefined geographic territories. Historically, this organizational form emerged in response to the high fixed costs of establishing distribution networks, including transportation costs and investments in packaging capacity. By contrast, a third upstream producer, UP3, does not operate a fully independent downstream system. Instead, it relies on third-party downstream firms, including participants in both Group 1 and Group 2, as well as other independent distributors, to produce and distribute its products. This arrangement allows UP3 to achieve broad geographic coverage without incurring the full costs of vertical integration.

Figure 1 illustrates this stylized vertical structure. UP1 and UP2 each maintain distinct downstream systems that do not overlap in their distribution of the same upstream products. UP3, in contrast, relies on shared downstream capacity drawn from both systems. This institutional environment generates heterogeneity in downstream incentives, as some downstream firms distribute products from a single upstream producer, while others distribute products from multiple upstream producers.

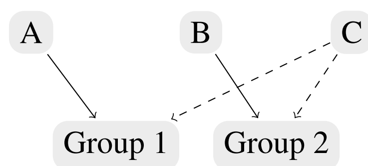
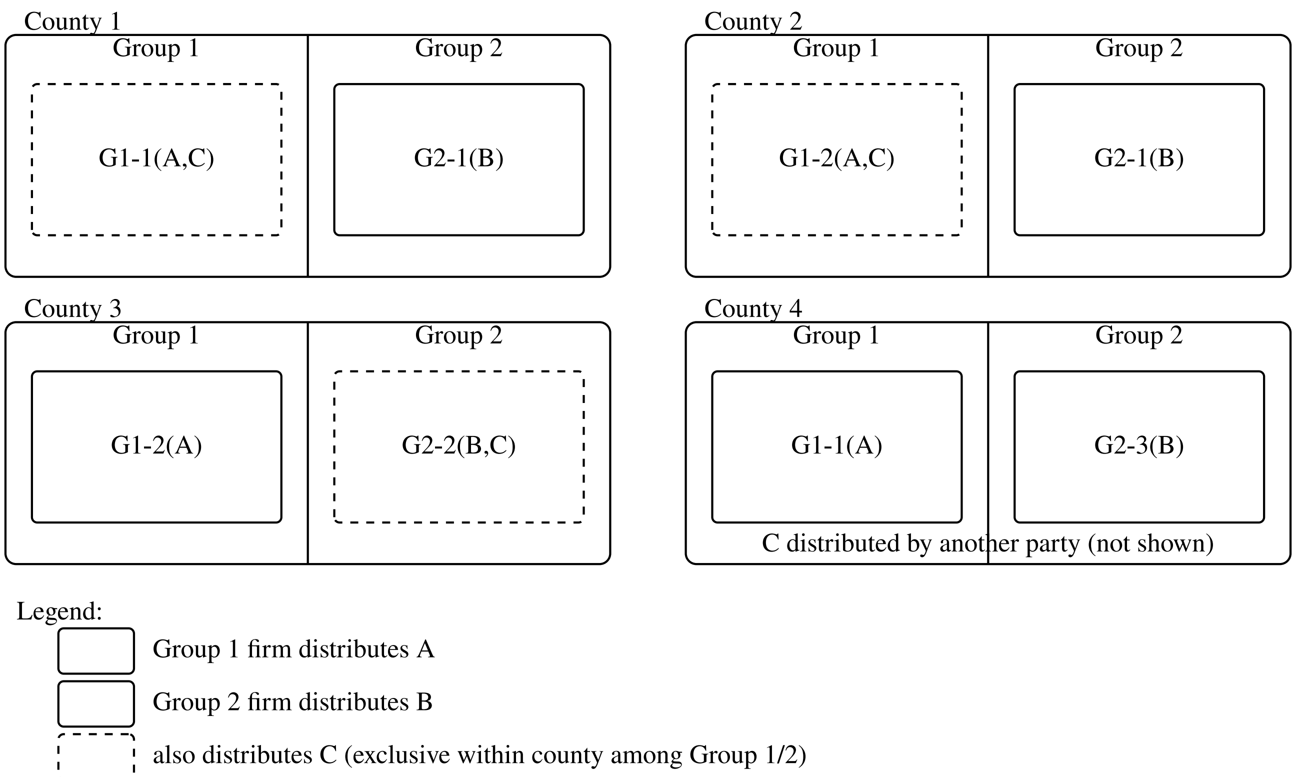


Figure 1: Stylized Vertical Structure in a Multi-Tier Supply Chain

Downstream distributors operate in exclusive geographic territories, as specified by contractual arrangements between upstream producers and downstream firms. These contracts grant downstream firms the sole right to distribute a given upstream producer's products within a defined area, often on a long-term or perpetual basis. Figure 2 provides a stylized illustration of exclusive downstream territories across counties within a Designated Media Market (DMA)[1]. In each county, exactly one downstream firm from Group 1 distributes products from UP1 (product A), and exactly one downstream firm from Group 2 distributes products from UP2 (product B). The identity of the downstream firm serving a given county may differ across counties, and the geographic partitions induced by Group 1 and Group 2 need not coincide.

UP3's products (product C) are distributed through third-party arrangements. In some counties, product C is distributed by a downstream firm affiliated with Group 1; in others, it is distributed by a downstream firm affiliated with Group 2; and in some

[1]Source: Nielsen Consumer LLC.



Notes: This schematic illustrates exclusivity within counties. Each county has one Group 1 distributor for A, and one Group 2 distributor for B. One of them may distribute C.
In some counties, C may be distributed by another party not shown in the figure.
Counties 1–2: Group 1 distributes both A and C. County 3: Group 2 distributes both B and C.
County 4: C is distributed by another party not shown.

Figure 2: Stylized Illustration of Exclusive Downstream Territories across Counties

counties, it is distributed by a separate downstream firm not shown in the figure. Within a given county, however, product C is carried by only one downstream firm. This structure implies that overlap between franchised products and third-party products occurs only in certain territories, creating variation in downstream firms' incentives.

Exclusive territories play a central role in the empirical analysis for two reasons. First, they allow sales volumes to be attributed to specific downstream firms at a fine geographic level using retail data. Second, they generate cross-county variation in exposure to vertical integration. Efficiency gains from integration arise in territories served by vertically integrated downstream firms, while foreclosure incentives arise only in counties where integrated downstream firms also distribute UP3's products. In contrast, when UP3 relies on downstream firms that are not vertically integrated with rival upstream producers, foreclosure concerns are attenuated. This geographic variation provides the basis for identifying the competitive effects of vertical integration in the analysis that follows.

2.2 Pricing Dynamics and Ownership Structure

Upstream producers charge downstream distributors a wholesale price for branded inputs. According to industry sources, wholesale prices reported by industry sources are typically expressed as annual prices per standardized unit, implying a linear pricing structure. Consistent with prior empirical work that documents limited use of nonlinear pricing in similar vertically structured consumer goods markets, I assume that upstream producers set linear wholesale prices that are uniform across downstream distributors within a given year (Luco and Marshall, 2020).

Downstream distributors, in turn, choose retail prices at the local market level while bearing responsibility for transportation, delivery, and in-store servicing of products. I abstract from strategic interactions between downstream distributors and retailers, treating downstream firms as the relevant pricing agents. This abstraction is appropriate given the institutional features of the distribution system, as well as data limitations: observed prices correspond to retail transaction prices, which reflect downstream pricing decisions but do not separately identify retailer margins.

Ownership relationships between upstream producers and downstream distributors further shape pricing incentives. Both UP1 and UP2 hold partial or full ownership stakes in a subset of their contracted downstream firms. Ownership data are obtained from industry sources described in Section 3. Prior to full vertical integration, UP1 and UP2 each held significant minority equity stakes in many downstream firms. Collectively, these partially owned downstream firms account for a large share of total distribution volume for both upstream producers—approximately 70 to 80 percent.

These ownership arrangements are particularly relevant because the same downstream firms also distribute products from UP3 and other upstream firms under third-party agreements. As a result, some downstream firms simultaneously distribute products from a partially aligned upstream owner and from an independent upstream rival. This structure implies that incentive alignment between upstream producers and key downstream firms may already be partially internalized even before full integration occurs, potentially attenuating efficiency gains from integration. At the same time, partial ownership provides upstream producers with influence over downstream firms that are critical to the distribution of rival products, which may facilitate foreclosure incentives in markets where distribution overlaps.

3 Data

This section describes the data used in the analysis. To evaluate the effects of vertical integration, I require information on (i) the geographic territories served by downstream distributors, (ii) retail prices and quantities sold within those territories, (iii) downstream cost shifters related to transportation, and (iv) upstream wholesale prices.

3.1 Distribution Territories of Major Downstream Firms

Identifying the geographic areas affected by vertical integration requires detailed information on the distribution territories served by downstream firms affiliated with UP1 and UP2, as well as the third-party arrangements used by UP3. To construct these territories, I use proprietary industry data obtained from a widely used publication that compiles information on downstream ownership links and exclusive territorial assignments.

The data cover distribution territories across 2,533 U.S. counties, representing the vast majority of downstream activity in the category. While territory boundaries vary across downstream firms, they remain largely stable over the study period. This stability allows the analysis to exploit cross-county variation in downstream affiliation while abstracting from endogenous territory reassignment.

These territory data are essential for identifying potential foreclosure effects. A substantial share of UP3's core product is distributed through downstream firms that are partially or fully owned by rival upstream producers. As a result, vertical integration may affect competitive outcomes primarily in counties where downstream distribution of rival products overlaps. The territorial structure therefore plays a central role in determining the magnitude and geographic scope of integration effects.

3.2 NielsenIQ Retail Scanner Data

To analyze downstream pricing behavior, I use retail transaction data from the NielsenIQ Retail Scanner dataset. The data contain weekly information on prices, quantities, and store characteristics collected from point-of-sale systems across more than 90 retail chains nationwide. The geographic coverage spans 2,533 counties in 48 continental states, corresponding to 205 Designated Media Areas (DMAs).[2]

Because downstream distributors operate in exclusive territories, retail scanner data can be used to attribute observed prices and quantities to specific downstream firms. I define a market as a county–quarter pair. Weekly data are aggregated to the quarterly level to mitigate potential stockpiling behavior (Hendel and Nevo, 2006). Market size is measured as total category sales within each county–quarter.

The analysis focuses on six flagship products offered by the three upstream producers: regular and sugar-free versions of brands X, Y, and Z. From this point forward, I adopt the notation X, Y, and Z for brands, while earlier sections use alternative labels for exposition. Products are aggregated from the UPC level to the brand–package–volume level using information on package size, quantity, and sugar-free indicators. This aggregation yields 46 distinct products. Prices are normalized per 100 ounces, a standard unit for comparing packaged products sold in different formats. Standard multi-unit packages account for the largest share of sales across all six brands, and larger packages are associated with lower unit prices. Including multiple package sizes is important for capturing substitution patterns both within and across brands, as well as for accounting for variation in production and distribution costs.

3.3 Wholesale Prices of High-Volume Products

To assess the extent to which vertical integration affects pricing through the elimination of double marginalization, it is necessary to distinguish between wholesale and retail margins. I obtain wholesale prices for the six high-volume products described above from a proprietary industry source. These prices are reported as annual prices per standardized unit (288 ounces) and are converted to prices per 100 ounces for consistency with the retail data.

[2] Source: Nielsen Consumer LLC.

Wholesale prices are reported to be uniform across downstream distributors within a given year and to change infrequently over time. Accordingly, I treat wholesale prices as common across markets within each period. While observed wholesale prices provide a useful benchmark, they may not fully capture effective transfer prices faced by downstream firms, particularly when ownership relationships are present. To allow for this possibility, the empirical model permits actual wholesale costs to differ proportionally from reported prices when decomposing downstream marginal costs. This approach provides flexibility to capture unobserved adjustments in upstream–downstream pricing while remaining consistent with the structure of the available data.

3.4 Downstream Production Locations

Transportation costs constitute an important component of downstream marginal costs under the direct-store delivery system. To proxy for these costs, I use information on the locations of downstream production facilities obtained from the same proprietary industry source described above.

I assume that products consumed in a given county are supplied from the geographically closest production facility operated by the downstream firm serving that county. For each county–facility pair, I calculate the distance between the centroid of the county where consumption occurs and the centroid of the county where the facility is located. Distances are computed using the NBER County Distance Database. Across all county–facility pairs in the sample, the average distance is 101.83 miles.

These distance measures enter the cost specification as shifters of downstream marginal costs. By exploiting geographic variation in distances across counties and downstream firms, the analysis captures heterogeneity in transportation costs that is independent of pricing incentives induced by vertical integration.

4 Evidence from Event Study

Prior empirical work studying vertical integration in differentiated consumer goods markets documents both efficiency gains and potential foreclosure effects on prices, though existing evidence typically abstracts from heterogeneity across integration events and downstream arrangements. For example, Luco and Marshall (2020) study pricing effects following vertical integration using retail scanner data and a reduced-form framework. While their analysis provides important evidence on average pricing responses, it does not distinguish across different integration events or downstream distribution environments, and it relies on a different dataset and geographic coverage.

To assess whether similar pricing patterns arise in my data, and to allow for heterogeneity across integration events and downstream arrangements, I implement an event-study design closely related to the approach used in that literature.

The data span a period during which two major vertical integration events occurred. I define treatment as the consummation of a vertical integration between an upstream producer and a set of downstream firms, and define the timing of treatment at the quarter in which the integration becomes effective. Because the sample includes both events, I estimate their effects separately.

Each observation corresponds to a product–county–quarter triple. For the first integration event (UP1), the treated group consists of observations in counties served by downstream firms in Group 1 that become vertically integrated. For the second integration event (UP2), the treated group consists of observations in counties served by the downstream firms in Group 2 that subsequently integrate. The control group consists of observations in counties served by downstream firms that remain unintegrated throughout the sample period.

Within both treated and control counties, I further distinguish whether products from UP3 are distributed through downstream firms affiliated with the integrating upstream producer or through other downstream firms. This distinction allows the analysis to separately identify changes in outcomes for integrated products and for third-party products distributed through integrated downstream firms.

I estimate the following difference-in-differences specification:

$$\begin{aligned} \log y_{jcq} &= \eta_1 Post_q + \eta_2 Treat_{jc} + \eta_3 Treat * Post_{jcq} \\ &+ \eta_4 ThirdParty * Treat * Post_{jcq} + c + FE_{DMA} + FE_{Quarter} + FE_j + \epsilon_{jcq} \end{aligned} \tag{1}$$

where $\log y_{jcq}$ denotes the outcome of interest for product j in county c and quarter q, which may be either the logarithm of price, $\log p_{jcq}$, or the logarithm of market share, $\log s_{jcq}$. $Post_q$ is an indicator equal to one in periods following the integration event. $Treat_{jc}$ equals one if product j is sold by an integrated downstream firm in county c. $ThirdParty_{jcq}$ equals one if product j is distributed through an integrated downstream firm under a third-party arrangement in county c and quarter q.

The specification includes DMA fixed effects,[3] quarter fixed effects, and product fixed effects. The error term ϵ_{jcq} captures unobserved shocks to prices or demand.

The parameters of interest are η_3 and η_4. The change in outcomes for integrated products in treated counties is given by $\eta_1 + \eta_3$, while the corresponding change in untreated counties is η_1. Thus, η_3 captures the impact of vertical integration on integrated products. For third-party products distributed through integrated downstream firms, the change in treated counties is $\eta_1 + \eta_3 + \eta_4$, while the change for non-piggybacking products in the same counties is $\eta_1 + \eta_3$. Accordingly, η_4 captures the differential effect of vertical integration on third-party products distributed by integrated downstream firms.

Tables 1 and 2 report the regression results for the two integration events. Across both events, prices of integrated products decline modestly and are statistically insignificant,

[3]Source: Nielsen Consumer LLC.

while prices of third-party products distributed through integrated downstream firms increase significantly. For example, brand Y products sold by integrated downstream firms experience price reductions of approximately 1.3 percent relative to comparable products sold by unintegrated downstream firms, while the corresponding magnitude for brand X is 0.69 percent. In contrast, prices of rival products (brand Z) distributed through integrated downstream firms increase by 6.6 percent in the first event and by 0.8 percent in the second event.

Consistent patterns also emerge in market shares. Following integration, integrated products gain market share, while third-party products distributed through integrated downstream firms lose market share. Relative to existing reduced-form studies, the qualitative implications and overall magnitudes are similar (Luco and Marshall, 2020; Adachi, 2020), though the estimated effect of the first integration event on the price of brand Z is larger in this sample.

Taken together, these reduced-form results provide evidence of both efficiency effects and foreclosure incentives associated with vertical integration. However, reduced-form estimates alone cannot disentangle the underlying mechanisms or account for the role of partial ownership. To address these issues, the next section develops a structural model of demand and supply that explicitly incorporates ownership links and downstream pricing incentives.

Table 1: Effects on Prices and Shares of UP1 Integration

	(1) Log Price	(2) Log Share
Post UP1 VI	-0.00307	-0.183^{***}
	(0.00288)	(0.00859)
UP1 Treat	-0.0934^{***}	-0.286^{***}
	(0.00245)	(0.00730)
UP1 Treat * Post	-0.0127^{***}	0.100^{***}
	(0.00282)	(0.00841)
ThirdParty * Treat * Post	0.0664^{***}	-0.532^{***}
	(0.00247)	(0.00736)
Constant	1.173^{***}	-5.165^{***}
	(0.00805)	(0.0240)
DMA FE	Yes	Yes
Quarter FE	Yes	Yes
Num. Obs	847253	847253

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

DMA source: Nielsen Consumer LLC

Table 2: Effects on Prices and Shares of UP2 Integration

	(1) Log Price	(2) Log Share
Post UP2 VI	0.000690	-0.193***
	(0.00282)	(0.00843)
UP2 Treat	0.0914***	0.356***
	(0.00196)	(0.00585)
UP2 Treat * Post	-0.00687***	0.0613***
	(0.00256)	(0.00763)
ThirdParty * Treat * Post	0.00801**	-0.366***
	(0.00317)	(0.00946)
Constant	1.121***	-5.354***
	(0.00800)	(0.0239)
DMA FE	Yes	Yes
Quarter FE	Yes	Yes
Num. Obs	847253	847253

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

DMA source: Nielsen Consumer LLC

5 Demand

5.1 Model and Estimation

A demand-side model is required to characterize consumer substitution patterns and to quantify how consumers respond to price changes induced by shifts in market structure. These demand responses are subsequently used to recover markups and marginal costs for downstream distributors and upstream producers in Section 6.

I estimate a nested logit demand model, which allows for flexible substitution patterns across differentiated products. This approach was introduced by Berry (1994) in the context of oligopoly markets with differentiated products, and later extended by Berry, Levinsohn and Pakes (1995) and Nevo (2000). I adopt a specification without random coefficients due to computational constraints.

The nested logit framework addresses price endogeneity and unobserved product characteristics under the assumption of Bertrand competition. By inverting market share equations, the model recovers implied mean utilities, which can be estimated using instrumental variables. Relative to the simple logit model, the nested logit allows consumers to substitute more flexibly toward similar products within predefined groups. Recent applications include Brenkers and Verboven (2006) and Miller and Weinberg (2017).

Assume that in market t, consumer i derives indirect utility from consuming product j according to Equation 2 (for simplicity, the market subscript t is omitted). In Equation 2, p_j denotes the price of product j, and x_j is a vector of observed product characteristics, including indicators for sugar-free versus regular formulation and for small versus family package size. Small or individual-sized products are defined as single-unit product no larger than one liter (33.8 oz), similar to Powell and Leider (2022). The term ξ_j captures unobserved product characteristics. The remaining terms represent idiosyncratic tastes, which are allowed to be correlated among products within the same nest h.

Cardell (1997) shows that if the group-level taste shock $\bar{\varepsilon}ih(j)$ follows an extreme value distribution, then the composite error term $\bar{\varepsilon}ih(j) + (1-\rho)\bar{\varepsilon}_{ij}$ is also extreme value distributed.

$$u_{ij} = \alpha p_j + \beta x_j + \xi_j + \bar{\varepsilon}ih(j) + (1-\rho)\bar{\varepsilon}ij. \tag{2}$$

I assume that the nesting structure is defined at the brand level. Each of the six brands constitutes a nest, and an additional nest represents the outside good ($j = 0$), which captures all other products not explicitly modeled. Figure 3 illustrates this structure, with seven nests at the upper level and multiple package–volume products within each nest. The parameter ρ governs the correlation of unobserved utility components within nests; as ρ approaches one, substitution within nests becomes stronger, while $\rho = 0$ corresponds to the simple logit model.

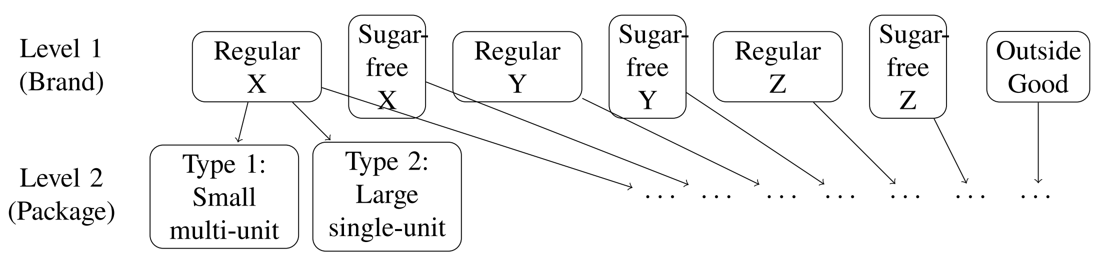


Figure 3: Nest Structure of the Demand Model

Consumers choose the product that maximizes indirect utility. Let J_h denote the set of products in nest h. The probability of choosing product j conditional on selecting nest h is given by

$$\bar{s}_{j|h} = \frac{exp\left(\frac{\delta_j}{1-\rho}\right)}{D_h}, \tag{3}$$

where

$$\delta_j = \alpha p_j + \beta x_j + \xi_j,$$

and

$$D_h = \sum_{k \in J_h} exp\left(\frac{\delta_k}{1-\rho}\right)$$

The probability that group h is selected is

$$\bar{s}_h = \frac{D_h^{1-\rho}}{\sum_h D_g^{1-\rho}}. \tag{4}$$

Thus the unconditional selection probability of product j in group h equals

$$s_j = \bar{s}_{j|h}\bar{s}_h = \frac{exp\left(\frac{\delta_j}{1-\rho}\right)}{D_h\left[\sum_h D_g^{1-\rho}\right]}. \tag{5}$$

Rearranging Equation 5, reinstating the market index t, and equating predicted choice probabilities with observed market shares yields the following estimable equation:

$$ln(s_{jt}) - ln(s_{0t}) = x_{jt}\beta - \alpha p_{jt} + (1-\rho)ln(s_{j|ht}) + \xi_{jt}. \tag{6}$$

Both prices p_{jt} and within-nest market shares $s_{j|ht}$ may be correlated with unobserved product characteristics. I therefore employ instrumental variables for both terms. Following common practice, the number of products available within each nest and market is used as an instrument for $s_{j|ht}$ (Brenkers and Verboven, 2006; Miller and Weinberg, 2017). This variable affects competitive conditions within the nest but is plausibly uncorrelated with product-specific unobservables, as brand formulations are fixed and variation arises from differences in package offerings across markets.

To instrument for prices, I use three sets of instruments. The first set consists of differentiation instruments following Gandhi and Houde (2019). These instruments exploit the proximity of products in characteristic space, which affects pricing incentives through competition but does not directly enter demand. Formally, for product j produced by firm f (so that $j \in J_{ft}$), the instruments take the form

$$Z_{jtl}^{\text{Local, Other}}(X) = \sum_{k \in J_{ft} \{ j} \mathbf{1}(|d_{jktl}| < SD_l),$$

$$Z_{jtl}^{\text{Local, Rival}}(X) = \sum_{k \notin J_{ft} \{ j} \mathbf{1}(|d_{jktl}| < SD_l),$$

where $d_{jktl} = x_{jlt} - x_{klt}$ denotes the difference in characteristic l between products j and k, SD_l is the standard deviation of these differences across markets, and $\mathbf{1}(\cdot)$ is an indicator function.

The second set of instruments exploits variation in vertical relationships: an indicator equal to one if a product is distributed through an integrated downstream firm following integration, and zero otherwise. This variable affects prices through supply-side incentives rather than consumer preferences. The third instrument is the distance (in thousands of miles) between the market and the closest production facility serving that market, which shifts downstream marginal costs under the direct-store delivery system. Table 3 reports the first-stage results, indicating that the instruments are relevant.

The parameters to be estimated are the price coefficient α, the characteristic coefficients β, and the nesting parameter ρ. Estimation proceeds via a two-step GMM procedure. In the first step, moment conditions exploit the orthogonality between the unobserved demand shocks ξ and the instruments. In the second step, the optimal weighting matrix is used. Estimation is implemented using the PyBLP package, which incorporates a range of best practices for demand estimation (Conlon and Gortmaker, 2020).

Table 3: First-Stage Results of Instruments

	(1) Prices	(2) Within-Group Shares
Differentiation IV 1	0.0299***	0.000414**
	(0.000352)	(0.000195)
Differentiation IV 2	0.00962***	-0.00355***
	(0.000342)	(0.000190)
Differentiation IV 3	0.0248***	-0.00101***
	(0.000339)	(0.000188)
Differentiation IV 4	0.0215***	-0.00272***
	(0.000352)	(0.000196)
VI Status	0.0504***	0.00582***
	(0.00159)	(0.000883)
Transportation Distance	0.000386***	-0.0000199***
	(0.00000787)	(0.00000437)
Num. Prod. in Nest	-0.0249***	-0.0553***
	(0.000655)	(0.000364)
small package	4.060***	-0.131***
	(0.00490)	(0.00272)
Sugar-free status	0.0228***	-0.00545***
	(0.00143)	(0.000794)
Constant	2.086***	0.833***
	(0.00362)	(0.00201)
Observations	847253	847253
R^2	0.886	0.072
F	734800.9	7319.7

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

5.2 Results

Table 4 reports the estimated demand parameters. The estimated price coefficient is negative, as expected, indicating that higher prices reduce consumer utility. The coefficients on observed product characteristics suggest that consumers derive higher utility from smaller, individual-sized packages relative to larger, family-sized packages, and lower utility from sugar-free variants relative to regular variants. The estimated nesting parameter is large, implying strong within-brand correlation in unobserved preferences. This result indicates that consumers exhibit substantial brand loyalty and tend to substitute primarily among different package formats within the same brand. Appendix A reports additional specifications and robustness checks that support the stability of these estimates.

Table 4: Demand Estimates

	Estimates
Prices	-1.4818
	(0.0104)
Small Package	6.6614
	(0.0465)
Sugar-free	-0.4578
	(0.0025)
Nesting Parameter	0.8252
	(0.0034)

Using the estimated demand parameters, I compute price elasticities of demand. Table 5 presents product-level elasticities for a commonly purchased multi-unit package, which accounts for a substantial share of sales across all brands. Each entry in the table represents the elasticity of demand for the row product with respect to a price change in the column product. For example, a one percent increase in the price of the sugar-free variant of brand X leads to a 12 percent decrease in its own quantity, while quantities of comparable products offered by other brands increase by approximately 0.13

percent. These elasticities reflect strong own-price sensitivity and limited substitution across brands at the product level.

Table 6 reports the corresponding diversion ratios for the same package format. The diversion ratio $\mathcal{D}_{jk}$ measures the fraction of consumers who switch to product k following a price increase for product j. Diagonal elements correspond to diversion to the outside good. For most products, approximately 18 percent of consumers substitute toward the outside option following a one percent price increase, while the corresponding diversion rate for regular brand X is approximately 27 percent. These results indicate that price changes induced by vertical integration may have meaningful effects on demand for products outside the set of focal brands, such as smaller brands or private-label alternatives (Lopez et al., 2015).

Table 5: Product-Level Elasticities: Representative Multi-Unit Package

Q\P	Sugar-free X	Regular X	Sugar-free Z	Regular Z	Sugar-free Y	Regular Y
Sugar-free X	-11.9638	0.1968	0.03	0.0645	0.0681	0.1405
Regular X	0.1334	-8.0598	0.03	0.0645	0.0681	0.1405
Sugar-free Z	0.1336	0.192	-12.1134	0.0646	0.0682	0.1407
Regular Z	0.1329	0.1924	0.03	-12.1406	0.0681	0.1406
Sugar-free Y	0.1343	0.1872	0.0294	0.0614	-11.6341	0.1406
Regular Y	0.1343	0.1871	0.0294	0.0615	0.0681	-11.2827

Table 6: Product-Level Diversion Ratios: Representative Multi-Unit Package

Q\P	Sugar-free X	Regular X	Sugar-free Z	Regular Z	Sugar-free Y	Regular Y
Sugar-free X	0.1861	0.0172	0.0026	0.0057	0.0062	0.0129
Regular X	0.0164	0.2687	0.0037	0.0081	0.0088	0.0182
Sugar-free Z	0.0112	0.0162	0.1817	0.0055	0.006	0.0125
Regular Z	0.0111	0.0161	0.0025	0.1826	0.006	0.0124
Sugar-free Y	0.0108	0.015	0.0024	0.005	0.1781	0.0119
Regular Y	0.0112	0.0154	0.0025	0.0052	0.0059	0.1835

$$\mathcal{D}_{jk} = -\frac{\partial s_{kt}}{\partial p_{jt}} / \frac{\partial s_{jt}}{\partial p_{jt}}$$

Table 7 reports elasticities across different package formats within the same brand for the sugar-free variant of brand X. Consistent with the high nesting parameter estimate, cross-price elasticities within the same brand are substantially larger than cross-brand elasticities reported in Table 5. This pattern further supports the conclusion that consumers primarily substitute among package formats within a brand rather than across brands.

To examine substitution patterns at a more aggregated level, Table 8 presents median brand-level price elasticities. These elasticities measure the percentage change in total quantities sold for the row brand following a one percent increase in prices across all package formats of the column brand. For instance, a one percent increase in prices for all packages of sugar-free X reduces its total quantity sold by 3.38 percent, while increasing quantities sold by other brands by approximately 28 percent.

Two key insights emerge from these results. First, all own-price elasticities are negative and exceed one in magnitude, indicating that consumers respond strongly to price changes by reallocating purchases across package formats within brands. Second, cross-brand elasticities are moderate but non-negligible, suggesting that price changes associated with efficiency gains or foreclosure effects may meaningfully affect substitution across brands. The magnitudes of these elasticities are comparable to, or slightly lower than, those reported in related studies that estimate demand in similar settings using retail scanner data.

Overall, the estimated demand system provides a detailed characterization of consumer substitution behavior that is central to the subsequent supply-side analysis. In particular, the elasticity and diversion estimates inform the recovery of marginal costs and markups for downstream distributors and upstream producers. In the next section,

Table 7: Product-Level Elasticities: Sugar-free X

Q\P	Format A (Multi-unit, small)	Format B (Single-unit, small)	Format C (Single-unit, large)	Format D (Multi-unit, small)	Format E (Multi-unit, small)	Format F (Multi-unit, small)	Format G (Multi-unit, medium)	Format H (Multi-unit, medium)	Format I (Multi-unit, small)
Format A (Multi-unit, small)	-11.9638	2.5948	2.8373	0.9102	1.044	1.6829	1.5717	1.055	0.9284
Format B (Single-unit, small)	8.663	-55.8583	2.8185	0.9108	1.0441	1.6829	1.5697	1.0549	0.9262
Format C (Single-unit, large)	8.6411	2.5935	-12.7252	0.9102	1.044	1.6829	1.5717	1.055	0.9281
Format D (Multi-unit, small)	9.0192	2.2429	2.3464	-19.2787	0.8612	1.6344	1.4904	1.0905	0.9636
Format E (Multi-unit, small)	8.954	2.1139	2.2348	0.6436	-18.2749	1.5716	1.5108	1.1101	0.93
Format F (Multi-unit, small)	7.8981	2.3931	2.9512	0.5658	0.167	-18.2264	1.283	0.0606	0.7769
Format G (Multi-unit, medium)	8.5878	2.4142	2.8636	1.1501	0.7208	1.7426	-23.4699	0.087	1.0298
Format H (Multi-unit, medium)	9.3137	2.0862	1.98	0.5146	1.4747	1.3773	1.5193	-19.5403	0.7984
Format I (Multi-unit, small)	8.8205	2.3453	2.6665	0.9026	1.0222	1.6774	1.5793	1.061	-30.2411

Notes: Formats A–I denote package formats within the same brand nest. Labels are anonymized; "small/medium/large" follow the size classification defined in Section 5.

Table 8: Brand-Level Price Elasticities of Demand

Q\P	Sugar-free X	Regular X	Sugar-free Z	Regular Z	Sugar-free Y	Regular Y
Sugar-free X	-3.3778	0.3256	0.0669	0.1402	0.1533	0.3067
Regular X	0.2898	-3.5678	0.0669	0.1403	0.1534	0.3067
Sugar-free Y	0.2839	0.3156	-3.6668	0.1464	0.1595	0.316
Regular Y	0.2839	0.3157	0.0694	-3.5633	0.1597	0.3161
Sugar-free Z	0.2787	0.3025	0.0668	0.1382	-3.3107	0.33
Regular Z	0.2788	0.3026	0.0668	0.1382	0.169	-3.0668

these demand estimates are combined with pricing and ownership information to evaluate the welfare effects of vertical integration under a series of counterfactual scenarios.

6 Supply

In this section, I develop a two-tier vertical model of downstream distributors and upstream producers. I derive the profit-maximizing conditions for downstream firms, recover their marginal costs of production, and estimate how these costs vary with vertical integration status. These results provide key inputs for analyzing upstream incentives and counterfactual market structures.

6.1 Downstream Distributors

I assume that downstream distributors engage in Nash–Bertrand competition, setting retail prices to maximize profits while taking upstream prices as given. Downstream firms do not internalize upstream producers' profits. Retail prices are assumed to be linear and expressed per 100 ounces for each product. Each downstream firm incurs a constant marginal cost mc^b_{jt} for product j in period t, which includes the wholesale price of the relevant upstream input as well as downstream costs such as transportation and packaging.

The profit of downstream firm b in period t is given by

$$\Pi^b t = \sum j \in \mathcal{J}^b t(pjt - mc^b_{jt}) s_{jt}(p), \tag{7}$$

where $\mathcal{J}^b_t$ denotes the set of products distributed by firm b, including both products supplied under contractual relationships with upstream producers and third-party products distributed under shared arrangements. This formulation implies that downstream firms internalize substitution patterns across all products they distribute. In particular, if vertical integration induces foreclosure effects on third-party products, downstream firms may partially offset lost sales by diverting demand toward their other products.

The first-order condition for profit maximization can be written in vector form as

$$p_t - mc^b_t = -\left(T_t * \Delta_{bt}\right)^{-1} s_t(p), \tag{8}$$

where T_t is the $|\mathcal{J}t| \times |\mathcal{J}t|$ ownership matrix at the downstream level, taking value one if the row and column products are distributed by the same downstream firm. The matrix Δbt captures consumer substitution patterns at the downstream level, with element (j, k) given by $\Delta j, k = \frac{\partial s_{kt}}{\partial p_{jt}}$.

Equation 8 allows the recovery of downstream price–cost margins and markups without directly observing wholesale prices (Berto Villas-Boas, 2007). The derivation is provided in Appendix B. This approach enables an examination of how downstream markups vary before and after vertical integration, as well as how integration affects rival downstream firms. Recovering marginal costs is important for two reasons. First, it allows the estimation of cost efficiencies associated with vertical integration. Second, holding cost parameters fixed in counterfactual simulations permits an evaluation of how equilibrium prices and quantities respond to alternative market structures.

Figure 4 displays the distributions of recovered marginal costs mc^b_t and markups (measured by the Lerner index) at the downstream level. The left panel shows marginal costs by package size. Smaller, individual-sized products exhibit higher marginal costs, averaging approximately \$6 per 100 ounces, while larger, family-sized products have substantially lower marginal costs, around \$1 per 100 ounces. The right panel shows that markups for smaller packages are approximately 10 percent, whereas larger packages exhibit markups of roughly 37 percent.

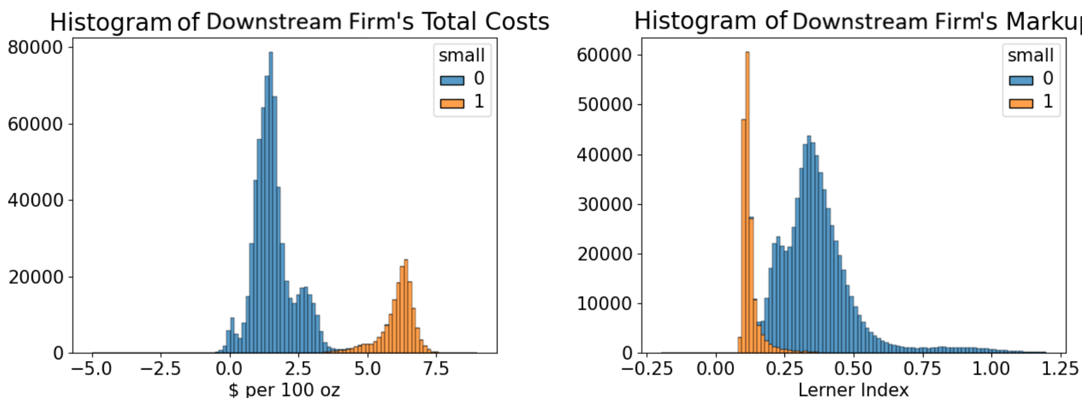


Figure 4: Distribution of Downstream Marginal Costs and Markups

To relate downstream marginal costs to vertical integration and other cost shifters, I parameterize mc^b_{jt} as

$$mc^b_{jt} = \lambda_1 p^f_t + \lambda_2 \text{VI}bjt + \lambda_3 \text{dist}bt + \sigma_{package} + \sigma_{quarter} + \eta_{bjt}, \tag{9}$$

where p^f_t denotes the reported wholesale price of the upstream input. The coefficient λ_1 allows effective wholesale costs to differ proportionally from reported prices due to unobserved contractual terms or measurement error. The coefficient λ_2 captures cost reductions associated with vertical integration, reflecting the elimination of upstream margins. The coefficient λ_3 measures the contribution of transportation costs to marginal costs, proxied by the distance between the downstream firm's nearest production facility and the market served. This specification is consistent with a direct-store delivery environment in which downstream firms bear transportation costs. The terms $\sigma_{package}$ and

$\sigma_{quarter}$ capture package-specific and time-specific fixed effects, and η_{bjt} is a structural error term.

Using marginal costs recovered from Equation 8, Table 9 reports estimates of Equation 13 obtained via OLS regressions by brand. Standard errors are computed using robust methods. Wholesale prices are positively correlated with marginal costs, as expected, though the estimated coefficients differ from one, potentially reflecting unobserved contractual arrangements or measurement error.

Integrated downstream firms associated with UP2 receive wholesale cost reductions of approximately 3 to 7 cents per 100 ounces, while integrated downstream firms associated with UP1 receive wholesale cost reductions of approximately 3 cents. These discounts correspond to reductions of roughly 21 to 72 percent of the upstream margin. The estimated magnitude of cost savings is consistent with prior empirical evidence on vertical integration, though somewhat smaller in size. Transportation distance is positively associated with marginal costs for most products, indicating the importance of delivery costs, though the estimated coefficient is negative for regular brand X.

Table 9: Downstream Marginal Cost Components

	(1) Sugar-free X	(2) Regular X	(3) Sugar-free Y	(4) Regular Y	(5) Sugar-free Z	(6) Regular Z
Wholesale Price	19.08***	161.3***	6.424***	10.10***	1.694***	3.725***
	(2.295)	(9.206)	(0.413)	(0.577)	(0.555)	(0.709)
VI	-0.0364***	-0.0779***	-0.0388***	-0.0355***		
	(0.00305)	(0.00408)	(0.00225)	(0.00202)		
Distance	0.000143***	-0.0000446**	0.000326***	0.000248***	0.000486***	0.000549***
	(0.0000158)	(0.0000206)	(0.0000117)	(0.0000105)	(0.0000108)	(0.00000983)
Quarter FE	Yes	Yes	Yes	Yes	Yes	Yes
Package FE	Yes	Yes	Yes	Yes	Yes	Yes
Num. Obs	162695	120744	142983	167500	114183	139148

Standard errors (in parentheses) are computed using the Huber/White/Sandwich estimator to be robust from heteroskedasticity of the errors.
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Having recovered downstream marginal costs and characterized how they vary with vertical integration, the next subsection examines the incentives of upstream producers. This step is necessary because upstream producers account for downstream pricing responses, and partial internalization of downstream profits may alter their incentives to vertically integrate.

6.2 Upstream Producers

This section uses the downstream results to estimate upstream producers' production costs, which are held fixed throughout the counterfactual analysis. I present a model in which upstream producers maximize profits not only from selling inputs to downstream firms but also by partially internalizing the profits of affiliated downstream distributors. These estimates form the basis for evaluating counterfactual market structures in the next section.

Assume that each upstream producer f competes by setting linear wholesale prices p^f (measured per 100 ounces) for its upstream inputs and has constant marginal costs of production, denoted mc^{f_r} for regular formulations and mc^{f_d} for sugar-free formulations. To remain consistent with the institutional environment described in Section 2, wholesale prices are assumed to be uniform across downstream firms for inputs of the same formulation type. As a result, each upstream producer sets two prices, p^{f_r} and p^{f_d}, in each quarter. The upstream producer's profit function is given by

$$\begin{aligned}\Pi_q^f = \sum_t M_t \Bigg\{ &(p^{f_r} - mc^{f_r}) \sum_{j \in \mathcal{J}_t^{f_r}} s_{jt}(p) + (p^{f_d} - mc^{f_d}) \sum_{j \in \mathcal{J}_t^{f_d}} s_{jt}(p) \\ &+ O_t^{fb} \times \sum_{j \in \mathcal{J}_t^b} \left[(p_{jt} - p_t^{f(j)} - mc_{jt}^b) s_{jt}(p) \right] \Bigg\} \end{aligned} \tag{10}$$

where t indexes counties within a given quarter. The set $\mathcal{J}_t^{f_r}$ includes products produced using upstream producer f's regular input, corresponding to different package formats of the same regular brand, while $\mathcal{J}_t^{f_d}$ includes products produced using its sugar-free input.

In each county, upstream producer f accounts for two components of profit. The first component captures profits from selling upstream inputs, aggregated over all downstream products that use those inputs. The second component captures the profits of affiliated downstream distributors, weighted by the equity ownership share $O_t^{fb} \in [0, 1]$. As discussed in Section 2, UP1 and UP2 partially internalize downstream profits through ownership links, while UP3 does not. Importantly, the downstream firm's product set $\mathcal{J}_t^b$ includes both the upstream producer's own branded products and third-party products distributed under shared arrangements. Internalizing downstream profits therefore affects incentives along two margins: it partially offsets double marginalization for the upstream producer's own products, while also creating incentives to influence prices of rival products in order to divert demand toward affiliated products. These forces interact with downstream pricing incentives and substitution patterns, as emphasized in the vertical pricing literature (Luco and Marshall, 2020).

Unlike the downstream pricing problem, the assumption that wholesale prices are uniform across downstream firms implies that upstream producers cannot solve their problem market by market. The first-order conditions with respect to p^{f_r} and p^{f_d} are given by

$$\begin{aligned} &\sum_t M_t \Bigg\{ \sum_{j \in \mathcal{J}_t^{f_r}} s_{jt} + (p^{f_d} - mc^{f_d}) \sum_{j \in \mathcal{J}_t^{f_r}} \frac{\partial s_{jt}}{\partial p^{f_r}} + (p^{f_r} - mc^{f_r}) \sum_{j \in \mathcal{J}_t^{f_d}} \frac{\partial s_{jt}}{\partial p^{f_r}} + \\ &+ O_t^{fb} \times \left[- \sum_{j \in \mathcal{J}_t^{f_r}} s_{jt}(p) + \sum_{j \in \mathcal{J}_t^{b}} \left(\frac{\partial s_{jt}(p)}{\partial p^{f_r}} (p_{jt} - p^{f(j)} - mc_{jt}^b) + \frac{\partial p_{jt}}{\partial p_{f_r}} s_{jt}(p) \right) \right] \Bigg\} = 0 \end{aligned} \tag{11}$$

$$\sum_t M_t \Bigg\{ \sum_{j \in \mathcal{J}_t^{f_d}} s_{jt} + (p^{f_d} - mc^{f_d}) \sum_{j \in \mathcal{J}_t^{f_r}} \frac{\partial s_{jt}}{\partial p^{f_d}} + (p^{f_r} - mc^{f_r}) \sum_{j \in \mathcal{J}_t^{f_d}} \frac{\partial s_{jt}}{\partial p^{f_d}} +$$

$$+ O_t^{fb} \times \left[- \sum_{j \in \mathcal{J}_t^{fr}} s_{jt}(p) + \sum_{j \in \mathcal{J}_t^{b}} \left(\frac{\partial s_{jt}(p)}{\partial p^{f_d}} (p_{jt} - p^{f(j)} - mc_{jt}^{b}) + \frac{\partial p_{jt}}{\partial p^{f_d}} s_{jt}(p) \right) \right] \Bigg\} = 0 \tag{12}$$

In both Equations 11 and 12, the first line captures how changes in upstream input prices affect profits from input sales. Increasing an input price raises margins on infra-marginal sales but reduces demand through higher downstream prices. The upstream producer also internalizes substitution toward its other formulation variant. The second line captures the effect on affiliated downstream profits, weighted by equity ownership. Because higher upstream prices raise downstream retail prices through pass-through, they may reduce downstream sales and margins. Partial internalization of these losses reduces incentives to raise wholesale prices, mitigating double marginalization. At the same time, the upstream producer internalizes the effect of price changes on third-party products distributed by affiliated downstream firms, which can create incentives to distort prices in order to divert demand toward its own products. The net effect depends on estimated demand elasticities from Section 5 and on upstream production costs, which are recovered by solving Equations 11 and 12.

Figure 5 reports estimates of upstream producers' marginal costs. Several patterns emerge. First, sugar-free formulations are more costly to produce than regular formulations. Second, production costs for brand Z remain relatively stable over the sample period, while brands X and Y exhibit noticeable increases following integration events. One interpretation is that greater internalization of downstream profits requires higher upstream marginal costs to rationalize observed prices. Third, brands X and Y exhibit lower marginal costs than brand Z, potentially reflecting differences in input technologies or scale economies.

Figure 6 presents estimated upstream markups. Markups for brand Z remain close to 60 percent throughout the sample. In contrast, brands X and Y exhibit high markups prior to integration that decline following integration events, consistent with improved alignment of incentives between upstream producers and downstream distributors. Markups charged to integrated and unintegrated downstream firms are similar, suggesting that the overall reduction in double marginalization at the upstream level is limited in magnitude.

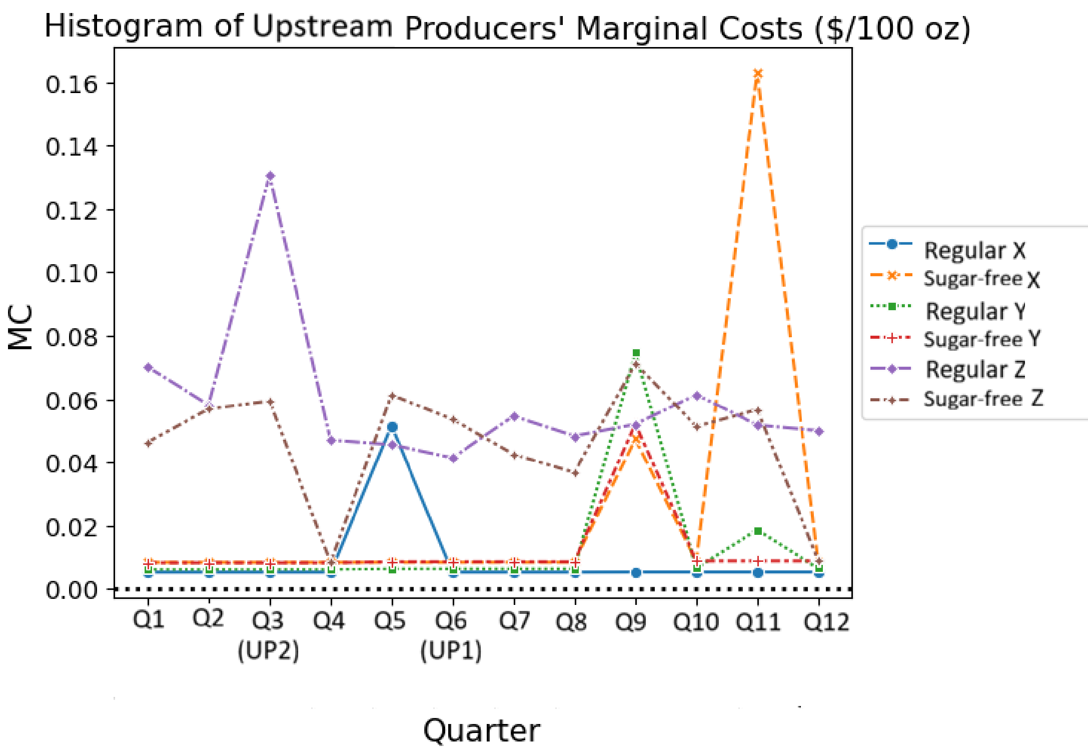


Figure 5: Estimates of Upstream Producers' Marginal Costs

Equipped with both demand- and supply-side estimates, the next section evaluates counterfactual scenarios corresponding to alternative vertical ownership and pricing structures.

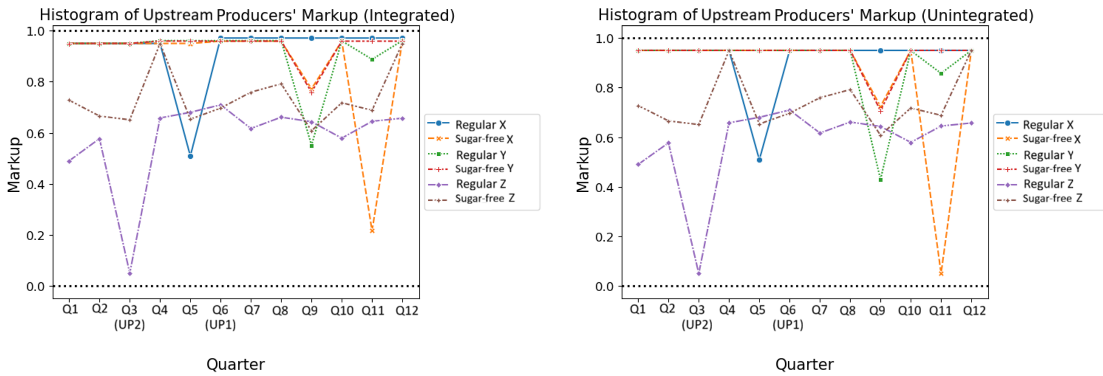


Figure 6: Estimates of the Upstream Producers' Markups

7 Counterfactual Exercises

The counterfactual exercises are designed to separate the efficiency and foreclosure channels associated with vertical integration in a two-tier vertical market with partial ownership and overlapping downstream distribution arrangements. I conduct three sets of counterfactuals.

The first two sets hold fixed the observed ownership links between upstream producers and downstream firms. In these counterfactuals, upstream producers continue to partially internalize affiliated downstream profits, as in the baseline equilibrium. The first counterfactual isolates the foreclosure channel by shutting down the cost-based efficiency component of integration: integrated downstream firms are assumed not to receive the integration-related marginal-cost reduction estimated from the cost decomposition. The second counterfactual isolates the efficiency channel by restricting the integrated upstream producer's ability to affect the retail prices of the third-party brand through downstream incentives, thereby eliminating the foreclosure mechanism while preserving cost efficiencies.

The third counterfactual considers a full divestiture scenario in which upstream producers do not internalize downstream profits. This scenario removes the ownership-based alignment of incentives and therefore eliminates both efficiency effects arising from internalization and foreclosure incentives operating through affiliated downstream firms. Comparing divestiture to the ownership-preserving counterfactuals quantifies the magnitude of these mechanisms under partial ownership.

Each counterfactual is computed using observed market fundamentals and follows a two-step procedure. The inner loop solves for downstream equilibrium prices and quantities given wholesale input prices. The outer loop solves upstream producers' pricing problems under the counterfactual market structure. In what follows, I report changes relative to the baseline (actual) equilibrium, defined as the difference between counterfactual and baseline outcomes, both in levels and as percentage changes, computed as (counterfactual − baseline)/baseline.

7.1 Counterfactual 1: Removing Efficiency

This set of counterfactual exercises isolates the efficiency component of vertical integration by assuming that integrated downstream firms do not receive the integration-related marginal-cost reduction estimated from Equation 13 (i.e., the term associated with λ_2 per 100 oz.). Importantly, ownership links are held fixed: upstream producers continue to partially internalize the profits of affiliated downstream firms as in the baseline equilibrium. The counterfactual therefore removes only the cost-based efficiency channel while preserving the incentive channel associated with partial ownership and downstream pricing.

I implement three related exercises: (1) removing the marginal-cost reduction associated with integration for both events, (2) removing it only for the event involving downstream firm B4, and (3) removing it only for the event involving downstream firms B1 and B2. For each exercise, I solve the counterfactual equilibrium prices and quantities and compute the resulting changes in downstream profits (by downstream firm and brand), upstream profits, and consumer surplus. All changes are reported as counterfactual minus baseline outcomes and are aggregated over the post-integration quarters.

$$mc_{jt}^{b} = \lambda_1 p_t^f + \lambda_2 \text{VI}bjt + \lambda_3 \text{dist}bt + \lambda_4 \text{Small}j + \sigma_b + \sigma_t + \eta bjt \tag{13}$$

7.1.1 Downstream Profit Changes

Figure 7 summarizes changes in downstream profits by brand across the three exercises. The first row reports the case in which efficiency effects are removed for both integration events, the second row removes efficiency only for the event involving B4, and the third row removes efficiency only for the event involving B1 and B2. Within each row, the left panel reports profit changes in millions of dollars and the right panel reports corresponding percentage changes.

In the first row, removing efficiency effects for both events yields substantial profit losses for integrated downstream firms in the upstream producers' focal brands. For example, in the scenario without efficiency gains from either event, B4's profits from brand X would be reduced by \$36M (9.492+26.777). B2's profits from brand Y would be reduced by approximately \$6M, and B1's profits from brand Y would be reduced by approximately \$17M, corresponding to reductions of roughly 2% to 6% across these outcomes. In contrast, profits from brand Z distributed by integrated downstream firms rise modestly, with increases ranging from 0.155 to 1.145 million dollars. This pattern reflects substitution effects: removing efficiency raises prices for the integrated producers' own brands, which can divert some demand toward the third-party brand within the same downstream firm's portfolio. Meanwhile, profits of unintegrated downstream firms change little, consistent with exclusive territories that limit within-brand competition in the same local market.

When removing efficiency effects only for the event involving B4, B4 experiences a decrease of approximately \$40M in profits from brand X, alongside an increase of roughly \$1M from brand Z. At the same time, B2 and B1 gain approximately \$1M and \$3M in profits from brand Y, respectively, relative to baseline, along with modest gains in brand Z. These spillovers are consistent with substitution between brands X and Y being stronger than substitution toward brand Z: weakening the cost advantage of brand X reallocates demand toward brand Y in overlapping territories and through competitive interactions across downstream firms.

Similarly, when removing efficiency effects only for the event involving B1 and B2, B1 and B2 earn approximately \$19M and \$7M less in profits from brand Y, respectively, compared with baseline. B4's profits increase by approximately \$3.6M. Profits from brand Z also increase slightly. Overall, these results indicate that the efficiency channel primarily operates through marginal-cost reductions for integrated downstream firms, and that the magnitude and distribution of profit impacts depend on downstream territorial overlap and cross-brand substitution.

7.1.2 Upstream Profit Changes

Table 10 reports changes in upstream producers' profits when efficiency effects are removed. In all three exercises, eliminating efficiency leads to higher upstream profits from selling wholesale inputs. This increase reflects higher wholesale prices in the absence of cost reductions passed through to downstream firms. The magnitude of the increase ranges between 18% and 26% across the different exercises.

These results highlight the classic tension between upstream and downstream incentives in vertically related markets. While efficiency gains from integration reduce downstream marginal costs and benefit consumers, they also compress upstream margins. Removing these efficiency gains therefore shifts surplus upstream by allowing producers to sustain higher wholesale prices.

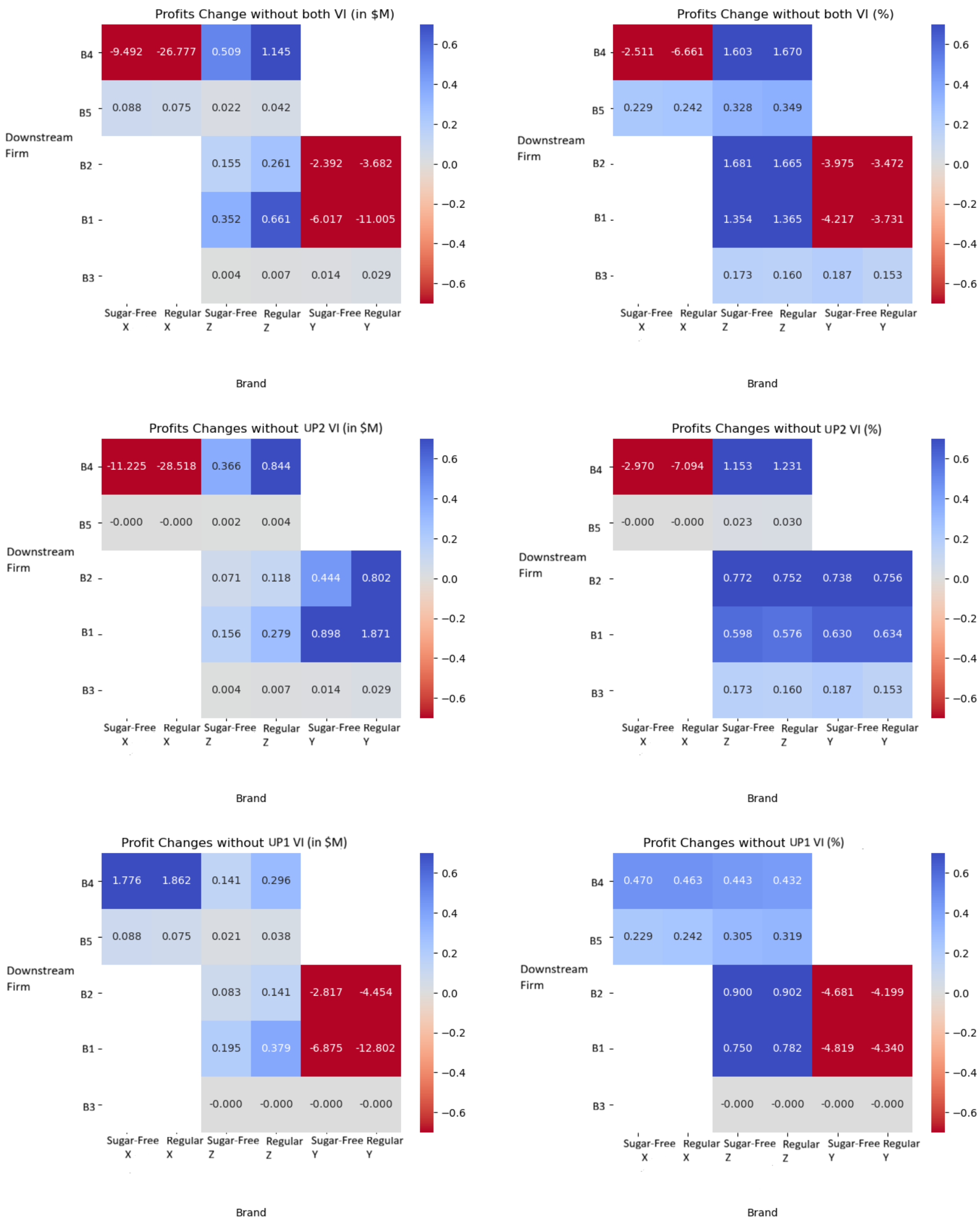


Figure 7: Downstream Profit Changes After Removing Efficiency

Table 10: Upstream Producer Profit Changes After Removing Efficiency

	Without Both VI		Without UP2 VI		Without UP1 VI	
Upstream Firm	Profit Δ ($M)	Profit Δ (%)	Profit Δ ($M)	Profit Δ (%)	Profit Δ ($M)	Profit Δ (%)
UP1	40.084	26.596	0	0	40.084	26.596
UP2	26.828	18.374	26.828	18.374	0	0
UP3	0	0	0	0	0	0

7.1.3 Consumer Surplus Changes

Table 11 presents changes in consumer surplus across the three exercises, aggregated by areas defined by overlapping downstream territories. When efficiency effects are removed for both integration events, consumers in areas jointly served by B4 and B1 experience a reduction in surplus of approximately $34M, corresponding to a 3.37% decline. Similarly, consumers in areas jointly served by B4 and B2 experience a decline of approximately $12.7M, or 3.34%. In areas served by downstream firms that remain unintegrated, the decrease in surplus ranges from 0.9% to 1.9%.

Aggregating across all markets, total consumer surplus declines by approximately $55M. Relative to the level of total surplus in the baseline equilibrium, this reduction is modest in percentage terms. These results indicate that efficiency gains from vertical integration are largely passed through to consumers in the form of lower retail prices.

When efficiency effects are removed for only one integration event, consumer losses are concentrated in areas served by the corresponding integrated downstream firms. For example, relative to the baseline equilibrium, eliminating efficiency effects associated with the event involving B4 leads to consumer surplus losses of approximately $27M in areas served by B4, contributing to a 1.1% reduction in total consumer surplus. Similarly, eliminating efficiency effects associated with the event involving B1 and B2 results in a 0.786% reduction in total consumer surplus. These patterns underscore the importance of downstream territorial overlap in shaping how efficiency gains are distributed across consumers.

Table 11: Consumer Surplus Changes After Removing Efficiency

	Without Both VI		Without UP2 VI		Without UP1 VI	
Downstream Territories	CS Δ ($M)	CS Δ (%)	CS Δ ($M)	CS Δ (%)	CS Δ ($M)	CS Δ (%)
B4 - B1	-33.974	-3.368	-20.398	-2.022	-13.542	-1.343
B4 - B2	-12.639	-3.347	-7.156	-1.895	-5.471	-1.449
B4 - B3	-0.232	-1.979	-0.232	-1.979	0	0
B5 - B1	-1.163	-1.433	0	0	-1.163	-1.433
B5 - B2	-0.012	-0.905	0	0	-0.012	-0.905
B5 - B3	0	0	0	0	0	0
Other	-7.044	-0.244	-4.503	-0.156	-2.541	-0.088
Total	-55.064	-1.906	-32.29	-1.118	-22.728	-0.786

7.1.4 Total Surplus Changes

Table 12 summarizes changes in total surplus under the counterfactual scenarios, while Table 13 reports more disaggregated results. When efficiency effects from both integration events are present in the baseline equilibrium, they contribute to a 0.687% increase in total surplus. Conversely, eliminating the efficiency effects of a single event reduces total surplus by 0.62% for the event involving B4 and by 0.062% for the event involving B1 and B2.

Taken together, these results indicate that efficiency gains from vertical integration increase total surplus, even though they redistribute surplus away from upstream producers toward downstream firms and consumers. Eliminating efficiency effects leads to lower downstream profits, higher upstream profits, and lower consumer surplus, with the losses outweighing the gains when evaluated in terms of total surplus.

Table 12: Total Surplus Changes After Removing Efficiency

	Actual ($M)	Without Both VI ($M)	Without UP2 VI ($M)	Without UP1 VI ($M)
	6115.339	6073.3	6077.438	6111.556
Total Surplus Δ (%)	/	-0.687	-0.62	-0.062

Table 13: Total Surplus Changes After Removing Efficiency (Disaggregated)

		Actual ($M)	Without Both VI ($M)	Without UP2 VI ($M)	Without UP1 VI ($M)
Downstream Firm	B1	705.851	689.842	709.055	686.747
	B2	259.739	254.081	261.174	252.693
	B3	46.064	46.117	46.117	46.064
	B4	1208.774	1174.16	1170.242	1212.849
	B5	123.702	123.931	123.708	123.925
	Other: UP1	221.561	222.25	222.25	221.561
	Other: UP2	164.772	165.177	164.772	165.177
	Other: UP3	143.817	144.889	144.577	144.125
	Total	2874.28	2820.447	2841.895	2853.141
Upstream Firm	UP1	146.013	172.841	172.841	146.013
	UP2	150.714	190.798	150.714	190.798
	UP3	52.196	52.196	52.196	52.196
	Total	348.923	415.835	375.751	389.007
Consumer		2892.136	2837.018	2859.792	2869.408
Total Surplus		6115.339	6073.3	6077.438	6111.556
Changes ($M)			-42.039	-37.901	-3.783
Changes (%)			-0.687	-0.62	-0.062

7.2 Counterfactual 2: Removing Foreclosure

Counterfactual 2 isolates the foreclosure channel of vertical integration by assuming that upstream producers no longer influence downstream pricing decisions for third-party products distributed through affiliated downstream firms. In this counterfactual, downstream firms set retail prices for brand Z independently, without accounting for the upstream producer's profits from its own brands when pricing third-party products. Ownership links and cost efficiencies are otherwise held fixed.

Absent foreclosure incentives, affiliated downstream firms may have weaker incentives to distort third-party prices upward in order to divert demand toward the upstream producer's own brands. As a result, prices of third-party products may fall relative to the baseline equilibrium.

To implement this counterfactual, I modify the upstream producers' first-order conditions in Equations 11 and 12 by excluding third-party products from the internalization of downstream profits. Specifically, the summation term is adjusted from $\sum_{j\in\mathcal{J}_t^b}$ to $\sum_{j\in\mathcal{J}_t^b\setminus\mathcal{J}_t^{SP3}}$, where $\mathcal{J}_t^{SP3}$ denotes the set of third-party (brand Z) products.

The pseudo-code for Counterfactual 2 is provided in Appendix C. For each candidate value of upstream wholesale prices, I recompute downstream marginal costs, equilibrium retail prices and quantities, and the associated substitution and pass-through matrices. These objects are then substituted into the modified upstream first-order conditions, and the wholesale prices that satisfy the conditions are selected.

7.2.1 Downstream Profit Changes

Figure 8 presents changes in downstream profits under Counterfactual 2 relative to the baseline equilibrium. When foreclosure incentives are removed, downstream firms earn higher profits from brand X, brand Z, and sugar-free brand Y. Among these, sugar-free brand Z exhibits the largest profit increase, ranging from 0.9% to 1%, followed by sugar-free brand Y at approximately 0.7%. Regular brand Z shows a modest increase of about 0.1%, while B4 experiences a slight decrease of 0.07%.

The downstream changes are smaller in magnitude than those observed for brand X products (approximately 0.4%) but larger than those for regular brand Y, which exhibits a 0.1% decrease. The pattern suggests that foreclosure constrained downstream firms' ability to extract profits from third-party products, particularly sugar-free brand Z. Once foreclosure incentives are removed, downstream firms are better able to profit from these products. At the same time, substitution patterns imply that downstream firms can also extract additional profits from brand X and sugar-free brand Y.

Overall, the magnitude of downstream profit changes is smaller than in Counterfactual 1 (Figure 7), indicating that the efficiency channel plays a more prominent role than foreclosure in shaping downstream profitability.

7.2.2 Upstream Profit Changes

The second column of Table 14 reports changes in upstream producers' profits from wholesale input sales after foreclosure incentives are removed. All upstream producers

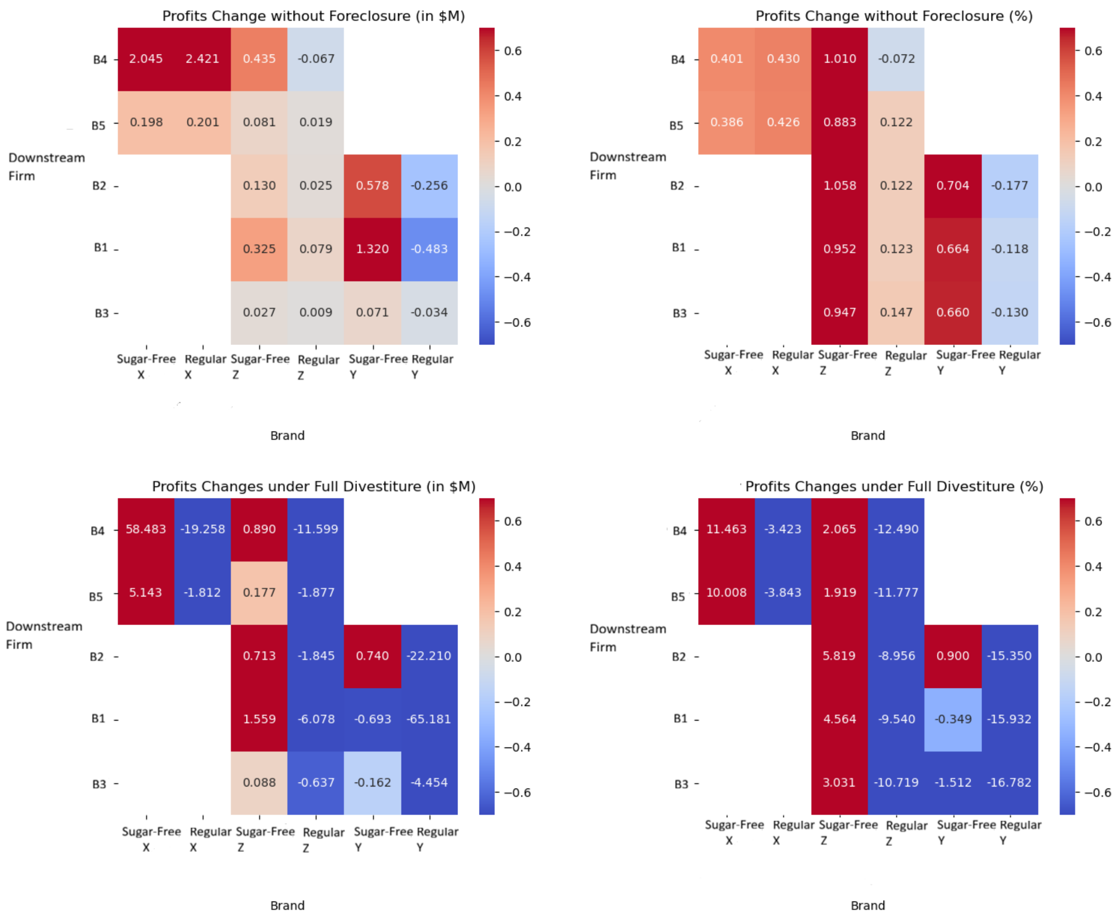


Figure 8: Downstream Profit Changes After Removing Foreclosure and under Full Divestiture

experience profit increases. The largest increase is observed for UP1 at 16.771%, followed by UP3 at 12.836% and UP2 at 8.509%.

Under foreclosure, upstream producers have incentives to lower wholesale prices in order to induce lower retail prices for their own products and gain a competitive advantage over third-party products distributed through affiliated downstream firms. This strategic pricing depresses equilibrium wholesale prices. When foreclosure incentives are removed, wholesale prices rise, and substitution patterns shift in a way that increases profits for all upstream producers, including UP3. These results underscore the importance of demand substitution patterns in determining how foreclosure affects upstream incentives.

Table 14: Upstream Producer Profit Changes After Removing Foreclosure and under Full Divestiture

	Without Foreclosure		Full Divestiture	
Upstream Firm	Profit Δ ($M)	Profit Δ (%)	Profit Δ ($M)	Profit Δ (%)
UP1	28.987	16.771	-17.181	-9.94
UP2	16.235	8.509	24.518	12.85
UP3	6.7	12.836	-13.903	-26.636

7.2.3 Consumer Surplus Changes

The left two columns of Table 15 report changes in consumer surplus, grouped by territories served by downstream firms affiliated with brand X and brand Y. Relative to the baseline equilibrium, removing foreclosure leads to consumer surplus losses ranging from 1.491% to 1.983%. The largest losses occur in territories served by integrated downstream firms (B1, B2, and B4).

Aggregating across all markets, removing foreclosure results in a 1.684% reduction in total consumer surplus, a magnitude similar to that observed when efficiency effects are eliminated. This implies that consumers are, on net, slightly better off under the baseline market structure. Although this result may appear counterintuitive, it is consistent with the producer-side outcomes. Foreclosure incentives can induce upstream producers and affiliated downstream firms to lower retail prices for both integrated and third-party products in certain markets, generating localized consumer benefits. In equilibrium, these strategic price distortions may translate into lower prices for some consumers.

Table 15: Consumer Surplus Changes After Removing Foreclosure and under Full Divestiture

	Without Foreclosure		Full Divestiture	
Downstream Territories	CS Δ ($M)	CS Δ (%)	CS Δ ($M)	CS Δ (%)
B4 - B1	-23.123	-1.665	-503.242	-36.247
B4 - B2	-8.229	-1.603	-182.406	-35.523
B4 - B3	-0.312	-1.887	-6.238	-37.779
B5 - B1	-1.959	-1.762	-42.094	-37.878
B5 - B2	-1.214	-1.983	-24.306	-39.712
B5 - B3	-0.027	-1.491	-0.619	-34.208
Other	-13.844	-1.743	-296.571	-37.34
Total	-48.707	-1.684	-1055.477	-36.49

7.2.4 Total Surplus Changes

Table 16 shows that when foreclosure incentives are removed, total surplus increases by 1.208%. This gain is primarily driven by higher profits for downstream firms and upstream producers, while consumer surplus declines slightly. The magnitude of the total surplus gain exceeds that observed in Counterfactual 1 (0.687%), indicating that the foreclosure channel plays a quantitatively important role in shaping overall market surplus. Table 17 reports more disaggregated results.

Table 16: Total Surplus Changes After Removing Foreclosure and under Full Divestiture

	Actual ($M)	Without Foreclosure ($M)	Full Divestiture
	6115.339	6189.2	5028.96
Total Surplus Δ (%)	/	1.208	-17.765

Table 17: Total Surplus Changes After Removing Foreclosure and under Full Divestiture (Disaggregated)

		Actual ($M)	Without Foreclosure ($M)	Full Divestiture ($M)
Downstream Firm	B1	705.851	707.091	635.458
	B2	259.739	260.217	237.137
	B3	46.064	46.136	40.899
	B4	1208.774	1213.609	1237.29
	B5	123.702	124.203	125.333
	Other: UP1	221.561	222.023	199.331
	Other: UP2	164.772	165.479	172.108
	Other: UP3	143.817	144.625	140.843
	Total	2874.28	2883.382	2788.399
Upstream Firm	UP1	146.013	201.828	155.66
	UP3	52.196	58.897	38.294
	UP2	150.714	207.033	215.317
	Total	348.923	467.758	409.271
Consumer		2892.136	2838.06	1831.29
Total Surplus		6115.339	6189.2	5028.96
Changes ($M)			6.949	-1153.291
Changes (%)			1.208	-17.765

7.3 Counterfactual 3: Full Divestiture

Counterfactual 3 evaluates market outcomes under full divestiture between upstream producers and downstream firms. This scenario removes all ownership-based links between the two layers, thereby eliminating efficiency gains from internalization, foreclosure incentives, and partial ownership effects. To implement this counterfactual, I restore double marginalization by imposing a common wholesale price across all downstream firms, regardless of prior integration status. In addition, I set $O^{fb} = 0$, so that upstream producers no longer internalize downstream profits.

While ownership links are removed, I retain downstream firms' existing product portfolios, including third-party products distributed under shared arrangements. This assumption reflects the limited ability of third-party producers to reassign distribution partners in the short run, given capacity constraints and territorial considerations.

The pseudo-code for Counterfactual 3 is provided in Appendix C.

7.3.1 Downstream Profit Changes

The bottom panel of Figure 8 reports changes in downstream profits under full divestiture. Relative to the baseline equilibrium, downstream firms affiliated with UP2 earn lower profits from regular products—approximately 3% for regular brand X and 12% for regular brand Z—but higher profits from sugar-free products, with increases of approximately 11% for sugar-free brand X and 2% for sugar-free brand Z.

Downstream firms affiliated with brand Y experience profit reductions from brand Y products, with declines of approximately 15% for regular brand Y and between 0.3% and 1.5% for sugar-free brand Y, with the exception of sugar-free brand Y distributed by B2. These firms also experience declines of 9–10% in profits from sugar-free brand Z, while profits from regular brand Z increase by approximately 3–5%.

Overall, these results suggest that under full divestiture, downstream firms extract relatively more profits from sugar-free products, whereas under the integrated structure, profits are more heavily concentrated in regular products.

One possible explanation is that vertical integration induces upstream producers to align wholesale pricing and incentives more closely with high-volume regular products, enhancing downstream profitability in those segments. Under divestiture, these strategic incentives disappear, allowing downstream firms to reallocate pricing power toward more differentiated segments, such as sugar-free products, where brand loyalty and reduced price sensitivity may support higher margins.

7.3.2 Upstream Profit Changes

The third column of Table 14 reports changes in upstream producers' profits from wholesale input sales under full divestiture. Relative to the baseline equilibrium, UP1 and UP3 experience profit reductions of approximately 10% and 27%, respectively, while UP2 experiences a profit increase of 12.85%.

These results indicate that full divestiture benefits only one upstream producer. Differences in brand strength, substitution patterns, and portfolio composition likely drive this asymmetry. UP2 may benefit from stronger consumer loyalty and lower cannibalization across its product line, whereas UP1 and UP3 appear to rely more heavily on coordinated pricing incentives that are disrupted under divestiture.

7.3.3 Consumer Surplus Changes

The rightmost two columns of Table 15 report changes in consumer surplus under full divestiture, grouped by territories associated with brand X and brand Y. In contrast to Counterfactuals 1 and 2, full divestiture leads to consumer surplus losses in all territories. Aggregating across markets, consumer surplus declines by 36.49%.

This substantial reduction indicates that the baseline market structure with partial ownership and vertical integration generates significantly higher consumer welfare than a fully independent upstream–downstream structure.

7.3.4 Total Surplus Changes

Table 16 summarizes changes in total surplus under full divestiture, while Table 17 reports disaggregated results. Relative to the baseline equilibrium, total surplus declines by 17.765%, driven primarily by large reductions in consumer surplus.

Taken together, these results suggest that full divestiture substantially worsens welfare outcomes. While it removes foreclosure incentives and restores double marginalization, it also eliminates beneficial coordination effects that lower prices and increase consumer surplus under the observed partially integrated structure.

8 Conclusions

This paper studies the competitive effects of vertical integration between upstream producers and downstream distributors in a differentiated-products market, with particular attention to foreclosure incentives affecting third-party brands. Combining reduced-form evidence with a structural demand-and-supply framework, the analysis estimates substitution patterns across brands and package formats and quantifies how prices, profits, and welfare respond to alternative vertical ownership structures.

While existing empirical work documents evidence consistent with both efficiency gains and foreclosure effects following vertical integration (Luco and Marshall, 2020; Adachi, 2020), this paper contributes by explicitly disentangling these mechanisms and tracing their welfare implications across upstream producers, downstream firms, and consumers. By constructing counterfactual scenarios that selectively remove efficiency, foreclosure, and ownership-based incentives, the analysis provides a more granular understanding of how vertical integration reshapes market outcomes.

The results reveal important asymmetries across mechanisms. Efficiency gains associated with reduced marginal costs increase total surplus by a modest 0.687%. In contrast, foreclosure incentives play a larger role in shaping aggregate welfare: eliminating foreclosure would raise total surplus by approximately 1.2%, driven primarily by higher profits for upstream producers and downstream firms, albeit accompanied by a small decline in consumer surplus. A full divestiture scenario yields markedly different outcomes, reducing total surplus by 17.765%, largely due to substantial losses in consumer surplus despite mixed effects on producer profits.

These findings highlight that vertical integration generates complex trade-offs. Efficiency gains improve total surplus but are quantitatively limited. Foreclosure incentives reduce total surplus yet may generate localized consumer benefits through strategic price distortions. Full separation of upstream and downstream firms eliminates both channels but results in large welfare losses, particularly for consumers.

Overall, the analysis underscores the importance of distinguishing between efficiency and foreclosure mechanisms when evaluating vertical integration. Policies that focus exclusively on price changes or average effects may overlook meaningful redistribution of surplus across market participants. The results suggest that careful scrutiny of foreclosure incentives in settings with partial ownership and overlapping downstream distribution is essential for understanding the full welfare consequences of vertical mergers.

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Appendices

A Additional Results using Alternative Demand Models

This appendix presents results from alternative demand model specifications. Table 18 reports estimates from a standard logit model and nested logit models with different nest definitions, and nested logit models with more granular packaging features.

Despite some differences in magnitude, the coefficients on price and the small-package indicator remain robust across specifications. The coefficient on the sugar-free indicator is also consistent in all models except the nested logit with upstream firms as the nest.

However, the nesting parameters in all alternative nested logit models reach their upper bound, suggesting that these nest definitions imply implausibly low substitution across groups. In contrast, the current specification yields more reasonable substitution patterns and is therefore preferred.

Due to data limitations in distribution territories for brands beyond the six primary ones studied, I am unable to report demand estimates with an expanded brand set. However, incorporating more brands could enhance the robustness of the demand estimation and remains an important direction for future analysis.

B Derivation of the Pass-through Rates

This appendix presents the derivation of the pass-through rate, building on the framework in Berto Villas-Boas (2007).

To obtain the pass-through rate Δ_{ft}, first note that $\Delta_{ft} = \Delta'_{pt}\Delta_{bt}$, where Δ_{pt} is a matrix of derivatives of all the retail prices with respect to all the wholesale prices, and Δ'_{bt} is the consumers' substitution matrix.

A key distinction is that in their setting, the number of products at the wholesale and

Table 18: Demand Parameters using Alternative Models

	Simple Logit	Nested Logit	Nested Logit	Nested Logit
Prices	-2.7447	-1.2705	-0.7565	-0.579
	(0.0156)	(0.0105)	(0.0076)	(0.0094)
Small	1.1885	5.8176	3.4944	/
	(0.0717)	(0.0465)	(0.0341)	
Sugar-free	-0.2909	0.0084	-0.4331	-0.5338
	(0.0043)	(0.0002)	(0.0014)	(0.0016)
Nest	/	Upstream Firm	Sugar-free	Brand
Nesting Parameter	/	0.99	0.99	0.99
		(0.0047)	(0.0025)	(0.0064)
Package FE	N	N	N	Y
Upstream Producer FE	Y	Y	Y	Y

retail levels is the same. In contrast, my setting involves six types of main ingredients at the wholesale level and up to 46 retail products, consisting of different packages of six brands. A change in the wholesale price of one ingredient affect all packages of that brand directly, and affects other products indirectly through consumer substitution. As a result, I modify both the derivation and the computational procedure to account for the mapping between wholesale and retail products.

Starting with the downstream producers' profit:

$$\Pi_t^b = \sum_{j \in \mathcal{J}_t^b} (p_{jt} - mc_{jt}^b) s_{jt}(p), \tag{14}$$

Take first-order condition with respect to p_{jt}:

$$s_{jt} + \sum_{m \in \mathcal{J}_t^b} \left[p_{mt} - p_{mt}^f - c_{mt}^b \right] \frac{\partial s_{mt}}{\partial p_{jt}} = 0 \quad \forall j \in \mathcal{J}_t^b. \tag{15}$$

Suppressing the subscript t for simpler presentation, taking total differentiation for a given j in 15 with respect to all prices ($dpk, \quad k = 1, \ldots, |\mathcal{J}^b|$) and a wholesale price

p_v^f, and denoting $|\mathcal{J}^b| = N$, we have

$$\sum_{k=1}^{N} \underbrace{\left[\frac{\partial s_j}{\partial p_k} + \sum_{i=1}^{N}\left(T(i,j)\frac{\partial^2 s_i}{\partial p_j \partial p_k}(p_i - p_i^f - c_i^b)\right) + T(k,j)\frac{\partial s_k}{\partial p_j}\right]}_{g(j,k)} dpk - \underbrace{T^{fb}(v,j)\frac{\partial s_v}{\partial p_j}}_{h(j,v)} dp_v^f = 0.$$

Let G be the matrix with element $g(j,k)$, H_v be row with general element $h(j,v)$, then $Gdp - H_v dp_v^f = 0$, so $\frac{dp}{dp_v^f} = G^{-1}H_v$. Stacking together all columns of H_v, we have $\Delta_p = G^{-1}H$, with general element $\Delta_p(i,j) = \frac{\partial p_j}{\partial p_i^f}$.

The derivation of G follows Berto Villas-Boas (2007), using the consumer substitution matrix, downstream price-cost margins, and the ownership matrix. In the expression for $H(j,v)$, the matrix T_{vj}^{fb} takes the value 1 if packaged product j is made from ingredient v, or if j is made from a different ingredient but its distributor also produces products made from v—either because they belong to the same upstream producer or due to third-party distributional arrangements across different upstream firms. Since there are six types of ingredients, the dimension of T^{fb} is $N \times 6$.

A change in the wholesale price p_v directly affects the shares of all products made with v. The aggregate share of these products is denoted s_v. Consequently, the dimension of H, and therefore Δ_p and Δ_f is also $N \times 6$.

C Pseudo Code for Counterfactuals II and III

This appendix presents the pseudo code for estimating Counterfactuals II and III in Section 1.7. The following pseudo code outlines the implementation.

Algorithm 1 Counterfactuals II & III

Input: starting values of upstream margin $margin_0^f$, upstream marginal costs mc^f, downstream marginal cost parameters λ, distance between downstream plants and consumers $dist$, ownership matrices T and T^{fb}, equity ownership matrix O^{fb}, market size M,

Output: Upstream margins $margin^f$

Outer loop:

Optimize over upstream margin $margin^f$ to minimize the difference between upstream firms' new FOCs and 0.

Inner loop:

for each candidate of $margin_{new}^f$ **do**
  for each market t **do**
    Feed mc^f
    if Counterfactual II **then**
      Feed λ, compute new wholesale prices for unintegrated downstream distributors and integrated downstream distributors $p_{new}^{f_u}$ and $p_{new}^{f_i}$ respectively
    end if
    if Counterfactual III **then**
      Compute new wholesale price p_{new}^f
    end if
    Feed λ and $dist$, compute new downstream marginal costs mc_{new}^b.
    Feed T, compute new equilibrium retail prices p_{new} and market shares s_{new}.
    Compute new Jacobian and Hessian matrices Jac_{new} and Hes_{new}, and downstream margin $margin_{new}^b$.
    Feed T^{fb}, compute pass-though matrix $\Delta_{p_{new}}$ and consumer substitution matrix with respect to wholesale prices $\Delta_{f_{new}}$.
  end for
  Feed upstream ownership matrix O^{fb} and market size M, aggregate across markets, compute the value of upstream producers' new FOCs.
  if Counterfactual II **then**
    Keep the original O^{fb}, replace $\sum_{j \in \mathcal{J}_t^b}$ with $\sum_{j \in \mathcal{J}_t^b \setminus \mathcal{J}_t^{SP3}}$
  end if
  if Counterfactual III **then**
    Set $O^{fb} = 0$
  end if
end for
Update $margin_{new}^f$.